

Commutative Algebra

Thobias Høivik

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Intro

These are my own personal notes in commutative algebra, a course I will hopefully be taking in the fall of 2026. Note that as these are for myself there might be gaps, inconsistencies and whatnot which might make it hard to follow so keep that in mind if you are considering using my notes for yourself.

1 Rings

Throughout the course, and these notes, a ring means a commutative ring with unity (meaning multiplicative identity). So every ring R has distinguished elements $1_R, 0_R \in R$ and multiplication satisfies $ab = ba$.

Definition 1.1 ((Commutative Unital) Ring). A ring is a set A together with binary operations $+, \cdot : A \times A \rightarrow A$ such that

1. $(A, +)$ is an abelian group with identity 0 ,
2. multiplication is associative,
3. multiplication is commutative,
4. multiplication distributes over addition,
5. there exists $1 \in A$ such that $1a = a$ for all $a \in A$.

Example 1.1. Examples of familiar rings include

$$\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$$

with the usual addition and multiplication.

If A is a ring, then the polynomial ring $A[x]$ is also a ring. More generally, $A[x_1, \dots, x_n]$ is the ring of polynomials in n commuting variables with coefficients in A .

Definition 1.2 (Ring homomorphism). Let A and B be rings. A function

$$\phi : A \rightarrow B$$

is a ring homomorphism if

1. $\phi(1_A) = 1_B$,
2. $\phi(a + a') = \phi(a) + \phi(a')$,
3. $\phi(aa') = \phi(a)\phi(a')$.

If ϕ is a bijection, then it is called an isomorphism.

The condition $\phi(1_A) = 1_B$ is a convention that one may or may not choose, but in this course it is not optional, and we say all homomorphisms must explicitly satisfy this.

Proposition 1.1. Let A be a ring. There exists a unique ring homomorphism

$$\iota: \mathbb{Z} \rightarrow A.$$

It is given by

$$\iota(n) = n \cdot 1_A.$$

Proof. Since ι is a ring homomorphism, it must satisfy

$$\iota(1) = 1_A.$$

Then additivity forces

$$\iota(2) = \iota(1 + 1) = \iota(1) + \iota(1) = 1_A + 1_A,$$

and similarly

$$\iota(n) = \sum_{k=1}^n 1_A = \underbrace{1_A + \cdots + 1_A}_{n \text{ times}}$$

for $n > 0$. Also,

$$\iota(-n) = -\iota(n).$$

This there is at most one such homomorphism. It is straightforward to check that the function $n \mapsto n \cdot 1_A$ preserves addition, multiplication, and sends 1 to 1_A . Hence it is a ring homomorphism. $\square$

Definition 1.3 (Characteristic). The characteristic of a ring A , denoted $\text{char}(A)$, is the unique integer $n \geq 0$ such that

$$\ker(\mathbb{Z} \rightarrow A) = (n).$$

Example 1.2. Here are the characteristics of a few familiar rings:

- $\text{char}(\mathbb{Z}) = 0$,
- $\text{char}(\mathbb{F}_p) = p$,
- $\text{char}(\mathbb{Q}) = 0$.

Intuitively, the characteristic is the smallest $n \in \mathbb{Z}^+$ such that $n \cdot 1_A = 0_A$ with $n := 0$ if no such n exists, and in fact, the characteristic is usually defined as such in a first course in abstract algebra.

Proposition 1.2 (Universal property of $\mathbb{Z}$ -polynomials). *Let A be a ring and let $a \in A$. There exists a unique ring homomorphism*

$$\phi_a : \mathbb{Z}[x] \rightarrow A$$

such that

$$\phi_a(x) = a.$$

It is given by

$$\phi_a \left(\sum_{k=0}^n c_k x^k \right) = c_0 \cdot 1_A + c_1 a + \cdots + c_n a^n.$$

Proof. A ring homomorphism $\phi : \mathbb{Z}[x] \rightarrow A$ must agree with the unique homomorphism $\mathbb{Z} \rightarrow A$ on constant polynomials.

If $\phi(x) = a$, then multiplicativity forces

$$\phi(x^n) = a^n.$$

Additivity then forces

$$\phi(c_0 + c_1 x + \cdots + c_n x^n) = c_0 \cdot 1_A + c_1 a + \cdots + c_n a^n.$$

Thus there is at most one such homomorphism.

Conversely, the formula above defines a ring homomorphism, by the usual rules for polynomial addition and multiplication. $\square$

So homomorphism $\mathbb{Z}[x] \rightarrow A$ are the same thing as choices of an element $a \in A$. Intuitively, $\mathbb{Z}[x]$ is the "freest" ring generated by one element.

Definition 1.4 (Ideal). *Let A be a ring. An ideal $I \subseteq A$ is a subset such that:*

1. *I is an additive subgroup of A ,*
2. *for every $a \in A$ and every $x \in I$, we have*

$$ax \in I.$$

Definition 1.5 (Principal ideal). *Let A be a ring and $a \in A$. The principal ideal by a is*

$$(a) = \{ra \mid r \in A\}.$$

An ideal is called principal if it is of the form (a) for some $a \in A$.

More generally, if $a_1, \dots, a_n \in A$, then

$$(a_1, \dots, a_n) = \{r_1 a_1 + \cdots + r_n a_n \mid r_i \in A\}.$$

This is the smallest ideal containing all of $a_1, \dots, a_n$.

Example 1.3. $(x, y) \subseteq k[x, y]$ is such an ideal.

$$(x, y) = \{f(x, y)x + g(x, y)y \mid f, g \in k[x, y]\}.$$

It is not the whole ring. It consists precisely of polynomials with zero constant term:

$$(x, y) = \{h \in k[x, y] \mid h(0, 0) = 0\}.$$

Example 1.4. $(0) = \{0\}$ is a (very uninteresting) ideal. $(1) = A$ is another (very uninteresting) ideal. For $n \in \mathbb{Z}$ we have that $(n) = n\mathbb{Z}$ is an ideal of $\mathbb{Z}$.

Proposition 1.3. Let $\phi: A \rightarrow B$ be a ring homomorphism. Then

$$\ker \phi = \{a \in A \mid \phi(a) = 0\}$$

is an ideal of A .

Proof. First we show that $\ker \phi$ is an additive subgroup. Since $\phi(0) = 0$, we have $0 \in \ker \phi$. If $x, y \in \ker \phi$, then

$$\phi(x - y) = \phi(x) - \phi(y) = 0 - 0 = 0.$$

Hence $x - y \in \ker \phi$. Now let $a \in A$ and $x \in \ker \phi$. Then

$$\phi(ax) = \phi(a)\phi(x) = \phi(a) \cdot 0 = 0.$$

Thus $ax \in \ker \phi$. Therefore $\ker \phi$ is an ideal. □

Definition 1.6 (Quotient ring). Let $I \subseteq A$ be an ideal. The quotient ring A/I is the set of additive cosets

$$a + I$$

with addition and multiplication defined by

$$(a + I) + (b + I) = (a + b) + I$$

and

$$(a + I)(b + I) = ab + I.$$

Proposition 1.4. If $I \subseteq A$ is an ideal, then A/I is a ring.

Proof. The quotient A/I is already an abelian group under addition because I is an additive subgroup.

We need to check multiplication is well-defined.

Suppose

$$a + I = a' + I$$

and

$$b + I = b' + I.$$

Then

$$a - a' \in I, \quad b - b' \in I.$$

We want to prove

$$ab + I = a'b' + I,$$

i.e.

$$ab - a'b' \in I.$$

Compute:

$$ab - a'b' = ab - a'b + a'b - a'b' = (a - a')b + a'(b - b').$$

Since $a - a' \in I$, and I absorbs multiplication by elements of A , we have

$$(a - a')b \in I.$$

Similarly,

$$a'(b - b') \in I.$$

Since I is closed under addition,

$$ab - a'b' \in I.$$

Therefore multiplication is well-defined.

The ring axioms for A/I then follow from the ring axioms in A . □

The quotient map is

$$\pi : A \rightarrow A/I, \quad a \mapsto a + I.$$

Its kernel is exactly I :

$$\ker \pi = I.$$

Theorem 1.1 (Fundamental homomorphism theorem). *Let*

$$\phi : A \rightarrow B$$

be a ring homomorphism. Then

$$\operatorname{im} \phi$$

is a subring of B , and there is an isomorphism

$$A/\ker \phi \cong \operatorname{im} \phi$$

given by

$$a + \ker \phi \mapsto \phi(a).$$

Proof. Define

$$\overline{\phi} : A/\ker \phi \rightarrow \operatorname{im} \phi$$

by

$$\overline{\phi}(a + \ker \phi) = \phi(a).$$

We first check well-definedness.

Suppose

$$a + \ker \phi = b + \ker \phi.$$

Then

$$a - b \in \ker \phi.$$

Thus

$$\phi(a - b) = 0.$$

So

$$\phi(a) - \phi(b) = 0,$$

and hence

$$\phi(a) = \phi(b).$$

Therefore $\overline{\phi}$ is well-defined.

It is a ring homomorphism because φ is.
 It is surjective onto $\text{im } \varphi$ by definition.
 Finally, its kernel is trivial. Indeed,

$$\overline{\varphi}(a + \ker \varphi) = 0$$

if and only if

$$\varphi(a) = 0,$$

if and only if

$$a \in \ker \varphi,$$

if and only if

$$a + \ker \varphi = 0 + \ker \varphi.$$

Thus $\overline{\varphi}$ is injective. Hence it is an isomorphism. $\square$

Example 1.5.

$$\mathbb{Z}[x]/(x^2 + 1) \cong \mathbb{Z}(i) = \{a + bi \mid a, b \in \mathbb{Z}\}$$

by $f \mapsto f(i)$.

We also have

$$k[x, y]/(x) \cong k(y)$$

by $f \mapsto f(0, y)$.

Lastly

$$k[x, y]/(x, y) \cong k$$

by $f \mapsto f(0, 0)$.

Theorem 1.2. Let $I \subseteq A$ be an ideal. There is a bijection

$$\{\text{ideals of } A/I\} \leftrightarrow \{\text{ideals } J \subseteq A \text{ such that } I \subseteq J\}.$$

The maps are

$$K \subseteq A/I \mapsto \pi^{-1}(K) \subseteq A,$$

and

$$J \subseteq A, I \subseteq J \mapsto J/I \subseteq A/I.$$

Proof. Let

$$\pi : A \rightarrow A/I$$

be the quotient map. First, if $K \subseteq A/I$ is an ideal, then $\pi^{-1}(K)$ is an ideal of A . Since $0 \in K$, and $\ker \pi = I$, we have

$$I \subseteq \pi^{-1}(K).$$

Conversely, suppose $J \subseteq A$ is an ideal with $I \subseteq J$. Then

$$J/I = \{j + I \mid j \in J\}$$

is an ideal of A/I . Now check that these constructions are inverse. If $K \subseteq A/I$, then

$$\pi(\pi^{-1}(K)) = K$$

because π is surjective. If $J \subseteq A$ contains I , then

$$\pi^{-1}(J/I) = J.$$

Indeed, $a \in \pi^{-1}(J/I)$ if and only if

$$a + I \in J/I,$$

which happens if and only if there exists $j \in J$ such that

$$a + I = j + I.$$

This means

$$a - j \in I.$$

Since $I \subseteq J$ and $j \in J$, we get $a \in J$. Hence

$$\pi^{-1}(J/I) = J.$$

Therefore the two maps are inverse bijections. □

Example 1.6. The ideals of $\mathbb{Z}/12\mathbb{Z}$ correspond to ideals of $\mathbb{Z}$ containing (12) . The ideals of $\mathbb{Z}$ are precisely (n) for $n \geq 0$. We need $(12) \subseteq (n)$. This happens precisely when $n \mid 12$. So the ideals of $\mathbb{Z}/12\mathbb{Z}$ are

$$(1), (2), (3), (4), (6), (12) = (0).$$

Proposition 1.5. Let k be a field. The ideal

$$(x, y) \subseteq k[x, y]$$

is not principal.

Proof. Suppose for a contradiction that

$$(x, y) = (f)$$

for some $f \in k[x, y]$. Since

$$x \in (x, y) = (f),$$

we have $f \mid x$. Since

$$y \in (x, y) = (f),$$

we have $f \mid y$. However, x and y have no common non-unit factor in $k[x, y]$. Hence f must be a unit. If f is a unit, then

$$(f) = (1) = k[x, y].$$

But this is impossible, because

$$1 \notin (x, y).$$

Therefore (x, y) is not principal. $\square$

Thus $k[x]$ is a principal ideal domain when k is a field, but $k[x, y]$ is not.

Exercises

Problem 1.1. Prove that if A is a ring with $1 = 0$, then A is the trivial ring.

Proof. Let $a \in A$. Then

$$\begin{aligned} a &= a \\ 1 \cdot a &= 0 \cdot a \\ a &= 0. \end{aligned}$$

Thus

$$A = \{0\}.$$

$\square$

Problem 1.2. Let A be a ring. Prove that there is a unique homomorphism $\mathbb{Z} \rightarrow A$.

Proof. Let A be a ring and let $\phi, \psi : \mathbb{Z} \rightarrow A$ be homomorphisms.

Then we require

$$\phi(1) = \psi(1) = 1_A.$$

Therefore, for $n > 0$, we have

$$\phi(n) = \phi\left(\sum_{k=1}^n 1\right) = \sum_{k=1}^n 1_A = \psi\left(\sum_{k=1}^n 1\right) = \psi(n).$$

For $n < 0$ we have $n = -m$ for some $m > 0$. Then

$$\phi(n) = \phi(-m) = -\phi(m) = -\sum_{k=1}^m 1_A = -\psi(m) = \psi(-m) = \psi(n).$$

Lastly,

$$\phi(0) = 0 = \psi(0).$$

Thus ϕ and ψ agree on all of $\mathbb{Z}$ and hence

$$\psi = \phi.$$

□

Problem 1.3. Identify $k[x, y]/(x, y - 1)$.

Solution. Let $\phi : k[x, y] \rightarrow k$ by

$$\phi(f) = f(0, 1).$$

Then ϕ is a ring homomorphism and furthermore it is surjective since, for $a \in k$ we have $f(x, y) := a \mapsto a$.

Now we show that $\ker \phi = (x, y - 1)$. First of all $x, y - 1 \in \ker \phi$ since $x(0, 1) = 0$ and $y - 1(0, 1) = 0$. Thus

$$(x, y - 1) \subseteq \ker \phi.$$

Now suppose $f \in \ker \phi$. Then $f(0, 1) = 0$ so $f(x, y) = xg(x, y) + (y - 1)h(x, y) + f(0, 1)$. Since $f(0, 1) = 0$ we have $f(x, y) = xg(x, y) + (y - 1)h(x, y)$ so $f(x, y) \in (x, y - 1)$. Thus $\ker \phi \subseteq (x, y - 1)$. The two together give $(x, y - 1) = \ker \phi$. Thus, by the fundamental theorem of ring homomorphisms (or first isomorphism theorem, if you prefer), we have

$$\frac{k[x, y]}{(x, y - 1)} \cong k.$$

□

Problem 1.4. List all ideals of $\mathbb{Z}/18\mathbb{Z}$.

Solution. The ideals are

- (0) ,
- (1) ,
- (2) ,
- (3) ,
- (6) ,
- (9) ,

i.e. (n) where $n \mid 18$. Note that $(0) = (18)$ and 18 divides 18.

□